Downloaded from http://sjg.lyellcollection.org/ at Purdue University Library on July 2, 2015

Palaeomagnetism of volcanics and sediments of the Eday Group, Southern Orkney

M. A. ROBINSON

Department of Earth Sciences, University of Leeds, Leeds LS2 9JT.

SYNOPSIS

This study presents the progressive demagnetisation results for 173 specimens from the Eday Group (upper Middle Old Red Sandstone), Orkney; including 85 from a contemporaneous lava and its baked contact, and 88 from Eday Group sediments. There is general agreement between stable NRM directions in the lava and its baked contact in the Eday Flags, which together give a mean direction after tectonic dip correction of $\bar{D} = 191°$, $\bar{I} = 41°$, corresponding to a palaeomagnetic pole at 8°N, 167°E. This may provide an approximate estimate of the upper Middle Devonian palaeomagnetic pole relative to Orkney.

All the palaeomagnetically stable sites elsewhere in the Eday Group sediments show a high blocking-temperature (T_B) component which give an *in situ* mean direction of $\bar{D} = 191°$, $\bar{I} = -20°$, corresponding to a palaeomagnetic pole position of 41°N, 163°E. This component is believed to be of secondary origin, and probably relates to a late Palaeozoic remagnetisation, as is apparently the case elsewhere in the Devonian of Europe and the USSR. However, there is insufficient between-site variation in tectonic dip to ascertain incontrovertibly the relative ages of remanence acquisition and deformation, which is generally considered to predate deposition of the Upper ORS.

A single site in the Eday Marls shows two poorly-defined ancient components, a low-T_B one in approximate agreement with the result from the lavas and a high-T_B component in a similar orientation to that found elsewhere in the Eday Group sediments. These are thought to correspond to an early (? detrital) remanence and late Palaeozoic remagnetisation respectively.

INTRODUCTION

The palaeomagnetic history of the British Devonian is very complicated and highly controversial, with conflicting views as to which data, if any, describe the Devonian palaeomagnetic field, and also with a number of proposed models for the timing and origin of the secondary magnetisations which are undoubtedly present; see Briden *et al.* (1973), Briden *et al.* (1984) and Tarling (this part) for a full discussion of this controversy. The picture is further complicated by the possible effects of large-scale transcurrent faulting within the British Isles, making palaeomagnetic correlation across such faults difficult to justify.

The aim of this paper is to present evidence of the local upper Middle Devonian

Downloaded from http://sjg.lyellcollection.org/ at Purdue University Library on July 2, 2015

palaeomagnetic field in Orkney by means of a consistency test between a contemporaneous lava and its baked contact in the Eday Flags, and then to use this result as a means to identify and distinguish any possible primary and secondary components within nearby sediments of the Eday Group.

The Eday Group sediments (Wilson *et al.* 1935; Mykura 1976) are essentially fine- to coarse-grained sandstones with interbedded flags in the lower part of the sequence and marls towards the top. The Lower Eday Sandstone, transitional from the Rousay Flags below, is a cross-bedded sandstone, yellow in the sampled area but reddening to the north (Astin, this part). This is overlain by drab, occasionally reddened, Eday Flags which are well-developed in the south but thin rapidly northwards. Within the Flags occur the lavas of Deerness, Shapinsay and Copinsay, extruded onto a wet surface and still warm when covered by further sediment (Wilson *et al.* 1935). The Middle Eday Sandstone above the flags is a cross-bedded red sandstone, thinning markedly to the north. It becomes marly towards the top, transitional to the red and green Eday Marls. The highest unit of the Eday Group is the red, pink and yellow Upper Eday Sandstone.

Deformation of the Middle Devonian is generally considered to predate deposition of the Upper Devonian (Wilson *et al.* 1935). The main feature is very open folding with an approximately N–S trend. This is more marked in the Northern Isles; the sampled area of Eastern Mainland and South Ronaldsay shows only limited tectonic disturbance, with dips no greater than 18°. Faulting occurs to a limited extent throughout the area, but there is no evidence of any large-scale strike-slip movement.

Sampling and treatment

Sampling was restricted to Eastern Mainland Orkney and the adjacent islands of Burray, S. Ronaldsay, Black Holm (Copinsay) and Shapinsay (Fig. 1). A total of 173 specimens from 21 sites have been progressively demagnetised, the majority, 148, by thermal demagnetisation while a further 25 were demagnetised by alternating field (A.F.). Of these 21 sites, 10 are from the lavas within the Eday Flags, including one site in the baked contact in the Eday Flags immediately below (Site 19). Eday Group sediments are represented by the Lower Eday Sandstone (Sites 11 to 13), Middle Eday Sandstone (Sites 16 to 18 and 24), Eday Marls (Sites 22, 30 and 31) and Upper Eday Sandstone (Site 21). Between 5 and 13 cores at each site were hand-drilled and individually oriented by means of a combination of sun (where possible) and magnetic bearings, giving rise to an orientation error of not greater than 2° (Collinson *et al.* 1967). From some of these cores, two specimens have been cut, in which case 'A' after the specimen number, eg. 2011A, denotes the upper core, nearer to the weathering surface.

Thermal demagnetisation was generally of between 10 and 15 steps, between 100°C and 600°–700°C, with a typical temperature interval of 80°–150°C at low

Downloaded from http://sjg.lyellcollection.org/ at Purdue University Library on July 2, 2015

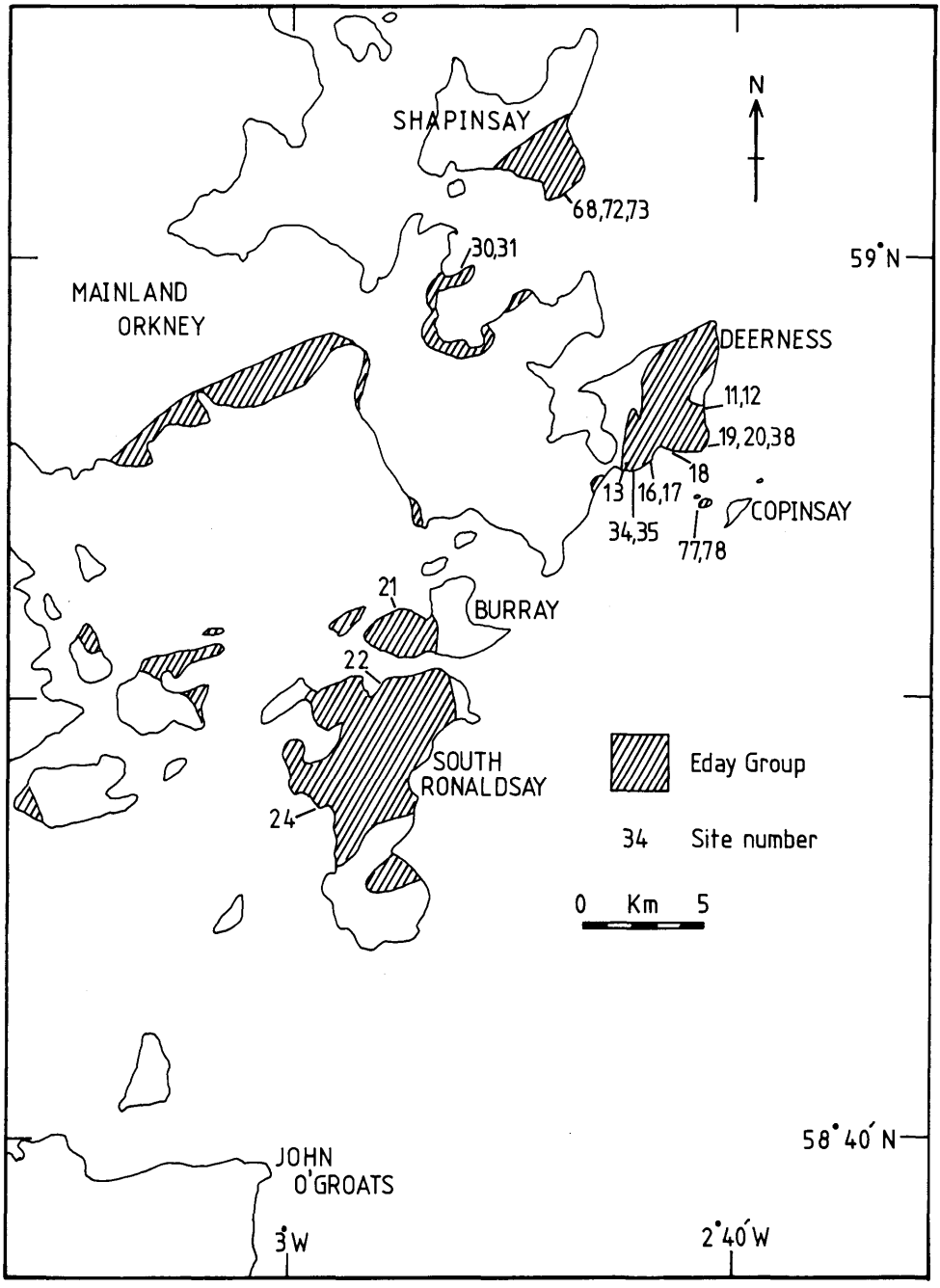


FIG. 1. Site locations in southern Orkney, showing the approximate outcrop area of the Eday Group.

Downloaded from http://sjg.lyellcollection.org/ at Purdue University Library on July 2, 2015

temperatures, and 10°–50°C above 500°C. A.F. demagnetisation of at least one specimen per site, in steps of between 2 and 20 mT up to a maximum of 190 mT was also used, but was generally found to be less effective than thermal treatment in the separation of multi-component remanences, perhaps due to the acquisition of spurious NRM components during demagnetisation, by grains having a low coercivity.

Specimens were measured on a CCL Cryogenic magnetometer (sediments and baked contact) or a Digico spinner magnetometer (lavas). Data are presented as orthogonal plots (Zijderfeld 1967) or stereographic projections. Individual components, both for single- and multi-component remanences, are estimated by use of the Linefind program (Kent *et al.* 1983).

Palaeomagnetism

Previous work

The lavas of Deerness have been investigated twice before. Morris *et al.* (1973) obtained a westerly, downwards result from A.F. cleaning of 9 specimens from Point of Ayre. This means of demagnetisation was found to be ineffective in the present study. Morris *et al.* (1973) included the result in estimating an overall mean direction from 7 sites from Hoy, Point of Ayre and Shetland, which is SSW and moderately steeply downwards (Table 1). This is somewhat at variance with the work of Storetvedt and Petersen (1972), working on the volcanic rocks of Hoy (Upper ORS) and Point of Ayre, who found a SSW, slightly downwards mean direction (Table 1). Their result was based on only 7 out of 60 measured samples and it was not stated which of these 7, if any, are from Point of Ayre. The pole position was, in the original author's opinion, considered 'somewhat inaccurate due to the problems of estimating single remanence components'.

The origin and timing of hematite genesis in the Orcadian Basin has been investigated by a number of workers; work up to 1976, summarised in Tarling *et al.* (1976), suggests that stable remanence-carrying hematite was not formed until the late Palaeozoic throughout most of the basin: see Tarling (this part) for an updated discussion of palaeomagnetic results from the basin.

Lavas and their contact zone in the Eday Flags

The nine sites in the Eday Lavas were collected over their entire outcrop range, from Copinsay in the south to Shapinsay in the north (Fig. 1). Progressive thermal demagnetisation generally results in one of two types of behaviour. The first type, typified by sample 3505A (Fig. 2a) shows a single-component magnetisation directed steeply downwards. Such a remanence does not correspond with any

Downloaded from http://sjg.lyellcollection.org/ at Purdue University Library on July 2, 2015

TABLE 1

Palaeomagnetic directions in Devonian rocks from the Orcadian Basin

Location/group	Dec	Inc	N	k	α_{95}	Palaeomagnetic pole position		Reference
Orkney Volcanics								
Orkney (Hoy, Eday)	205·0	+8·4	7[(1)]	44·8	9·1°	23·7°N,	149·5°E	1
Orkney (Eday)★	273·0	+67·0	9[(1)]	16·7		42·1°S,	116·7°E	2
Orkney & Shetland	216·0	+46·0	7[(2)]	18·4		2°S,	147°E	2
Orkney (Eday)[(a)]	191·3	+40·8	45[(1)]	7·5	8·3°	7·5°N,	166·6°E	
Orkney (Eday)[(a)]	193·8	+40·8	6[(2)]	20	15·2°	7·3°N,	164·3°E	3
Probable primary remanences in sediments								
Gamrie Outlier	184	+42			20°	8°N,	173°E	4
Eday Marls[(a)]	210·4	+74·1	5[(1)]	30·1	14·1°	31·6°S,	160·2°E	3
Hematitic remanences in sediments								
Lower-Middle Devonian	190	−39	12[(2)]	11	15°	53°N,	160°E	5
Lower Devonian	205·0	+3·1	27[(1)]	20	6·4°	27·0N,	148·5°E	6
Gamrie Outlier	214	−40			17°	47°N,	127°E	4
John O'Groats Sandstone	188·3	−41·1	43[(1)]	32	3·9°	54·2°N,	163·6°N	7
Eday Group[(b)]	190·7	−19·9	38[(1)]	16·7	5·9	40·6°N,	163·2°E	3
Eday Group[(b)]	190·9	−18·3	6[(2)]	31·4	12·1°	39·7°N,	163·0°N	

[(1)] Unit weight to samples
[(2)] Unit weight to sites
[(a)] Dip corrected
[(b)] *In situ*
★ Included in the Orkney and Shetland result below

References
1 = Storetvedt and Petersen (1972): Thermal demagnetisation, selected samples.
2 = Morris *et al.* (1973): A.F. cleaning.
3 = This study: Progressive thermal demagnetisation.
4 = Turner and Archer (1975): Thermal cleaning.
5 = Tarling *et al.* (1976): Thermal cleaning.
6 = Storetvedt and Torsvik (1983): Progressive thermal demagnetisation.
7 = Storetvedt and Carmichael (1979): Progressive demagnetisation (thermal and chemical).

known European Phanerozoic direction. The origin of these components remains somewhat obscure, with a number of possible explanations such as a drilling-induced magnetisation. They are not considered to produce a geologically significant direction. Such sites include the two on Copinsay, one on Shipinsay and one in Deerness.

The second type of behaviour (e.g. 7301A from Shapinsay; 2010 from Deerness; Fig. 2b, c) usually gives a low blocking temperature (T_B) component close to the present earth's field (P.E.F.) direction and a higher-T_B component directed downwards to the south. There is no stable magnetisation above 585°C. NRM intensities vary between 120 and 620 mA m^{-1}. Five specimens from this group were A.F. demagnetised, but this was found to be an ineffective method of

Downloaded from http://sjg.lyellcollection.org/ at Purdue University Library on July 2, 2015

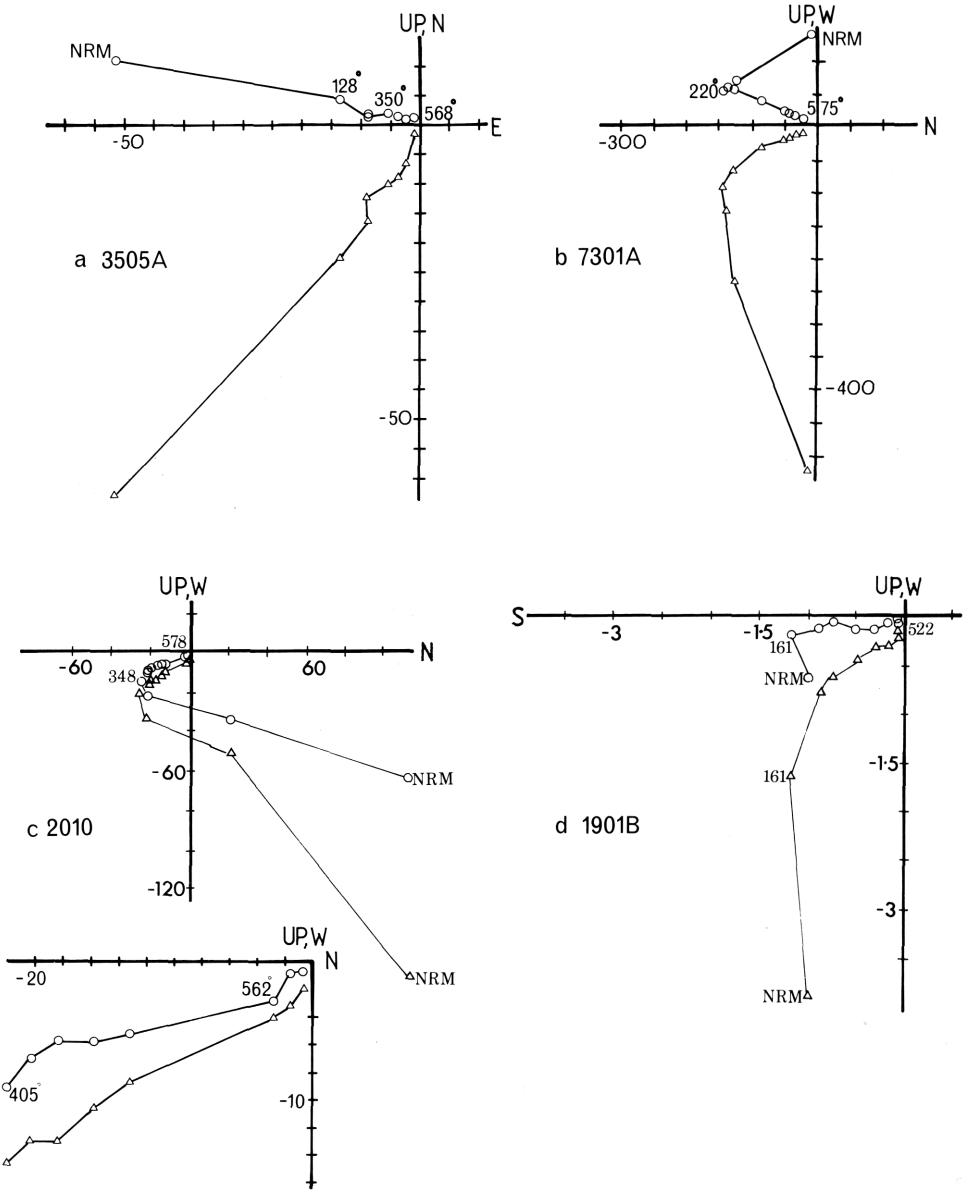


FIG. 2. Representative examples of thermal demagnetisation behaviour from Eday group lavas and sediments, presented as orthogonal projections (Zijderfeld 1967). Circles represent points in the horizontal plane, triangles those in the vertical plane. Peak demagnetisation temperatures in °C; Intensities in mA m^{-1}. All are presented *in situ*. (a) 3505A, Eday Lavas. Recently-acquired single component remanence. (b) 7301A, Eday Lavas. Low-T_B P.E.F. component, high-T_B component directed downwards to the south. (c) 2010, Eday Lavas. As previous example; inset shows high-temperature behaviour. (d) 1901B contact zone, Eday Flags. Southerly, downwards up to 522°C;

Downloaded from http://sjg.lyellcollection.org/ at Purdue University Library on July 2, 2015

separating NRM components and the results are not considered further. Other samples excluded from mean directions are six which have a much lower NRM intensity than the remainder (2–40 mA m^{-1}) and are magnetised parallel to the P.E.F. direction, two other strongly-magnetised samples with a P.E.F. direction, and three which give a magnetic orientation very different to the remainder, suggesting misorientation during collection.

Mean directions from the 5 acceptable sites (45 samples) are given in Figure 3a and Table 2. Within-site distribution is sometimes poor, especially site 20 (k = 6.0) which shows a distribution elongated along a NW–SE plane. Such a distribution does not occur in other sites which are generally of higher precision.

Also included in Figure 3a is the mean direction from site 19, collected in the contact zone in the Eday Flags immediately beneath the lavas of site 20, at distances

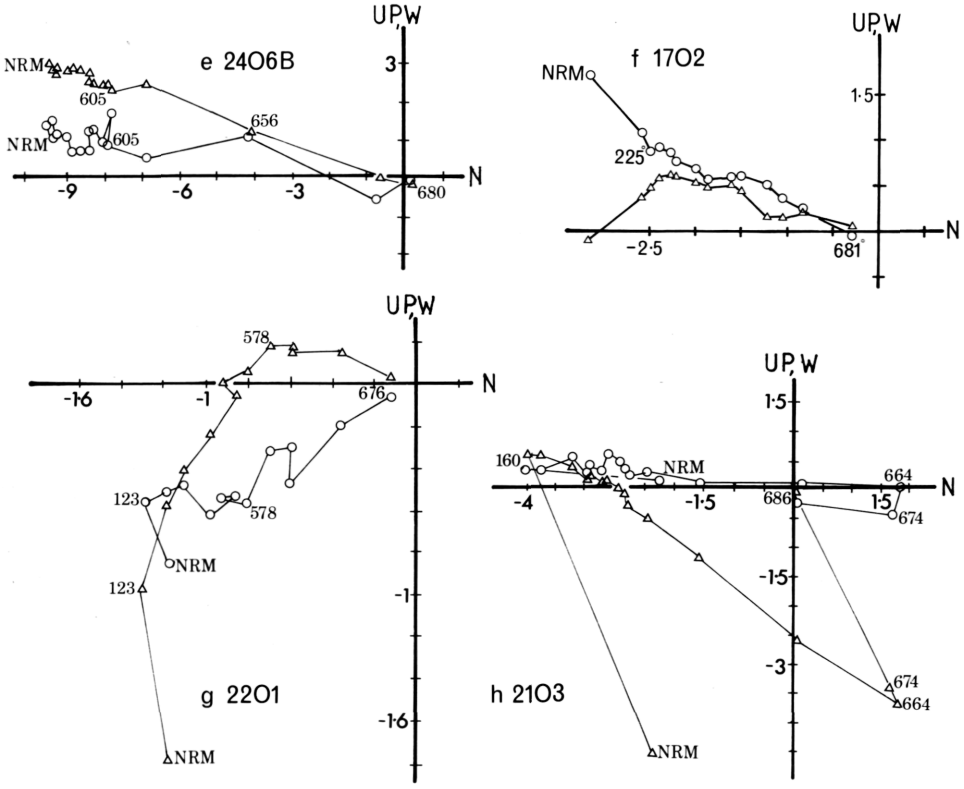


FIG. 2. (*continued*)
chemical alteration occurs above this temperature. (e) 2406B, Middle Eday Sandstone. Single-component, southerly and upwards. (f) 1702, Middle Eday Sandstone. Soft, recent component followed by southerly, slightly upwards. (g) 2201, Eday Marls. Southerly, downwards component between 123° and 578°, with a rapid inclination change to slightly upwards above 578°C. (h) 2103, Upper Eday Sandstone. P.E.F. components at both low T_B (NRM-160°C) and high-T_B (664–686°C). Intermediate component southerly and upwards.

Downloaded from http://sjg.lyellcollection.org/ at Purdue University Library on July 2, 2015

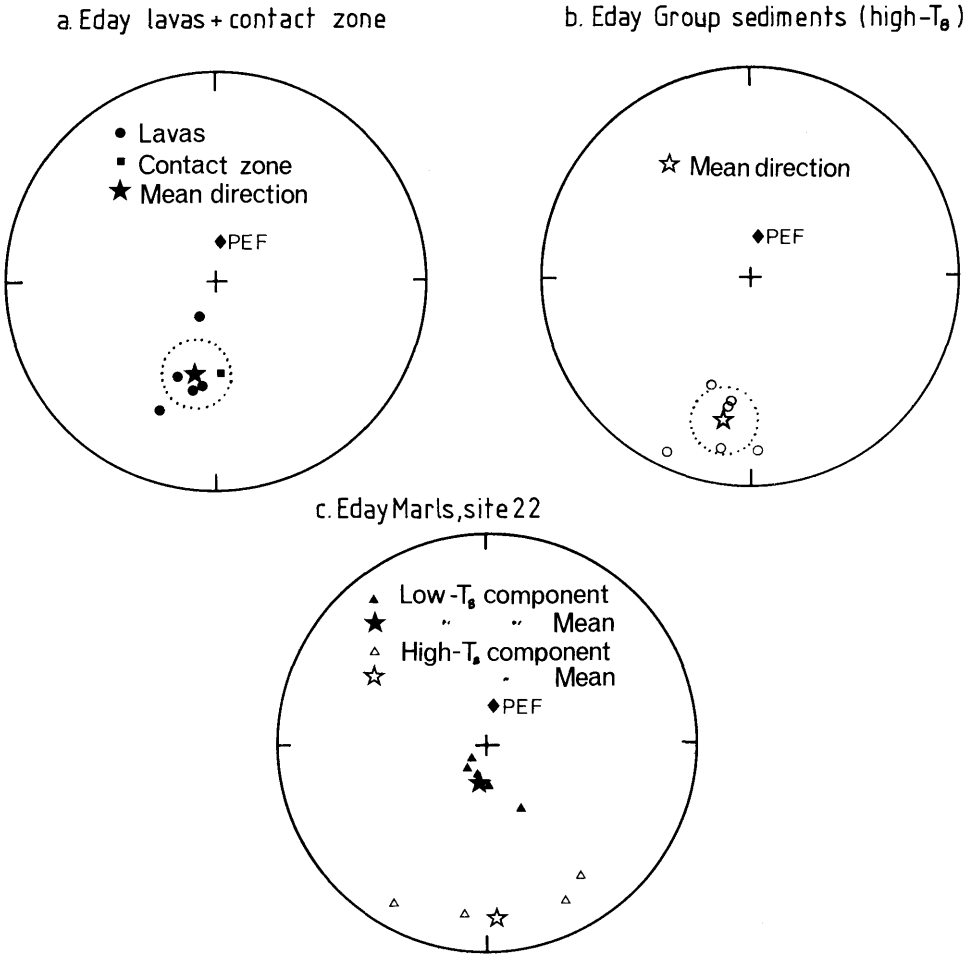


FIG. 3. Stable component distributions. Solid symbols represent directions in the lower hemisphere of the stereographic projection. (a) Lavas and contact zone site means dip corrected; (b) Eday Group sediments, high-T_B component site means (see Table 2), *in situ*; (c) Site 22, Eday Marls. Shows both low-T_B and high-T_B components, at sample level.

of between 0·01 m and 1·20 m beneath the contact. Typically they again show a low-T_B component close to the P.E.F. and a higher-T_B component directed downwards to the south (e.g. 1901B, Fig. 2d). On demagnetisation at temperatures above 550–600°C, most specimens become magnetically unstable with a tendency to pick up a strong, viscous component which obscures any further components that may have survived. NRM intensities vary between 2 and 12 mA m^{-1}, which is lower than that of the overlying lavas by an order of magnitude or more. The mean direction from this site is statistically similar to that

TABLE 2

Site mean directions

Site No	Formation	Location	In Situ Dec°	In Situ Inc°	Tectonic Correction[1]	Dip Corrected Dec°	Dip Corrected Inc°	N	k	α_{95}
			Components with T_B max ≤ 585°C							
19	Contact zone	Deerness	173·7	51·8	10°/090	174·8	41·8	11	21·3	10·1°
20	Eday Lavas	Deerness	192·5	44·0	10°/090	190·8	34·2	8	6·0	24·8°
34	Eday Lavas	Deerness	224·8	53·5	26°/334	201·5	69·7	8	3·8	33·1°
38	Eday Lavas	Deerness	189·5	45·3	10°/090	188·1	35·4	7	20·5	13·7°
68	Eday Lavas	Shapinsay	191·0	37·4	14°/200	201·4	38·3	4	11·9	27·8°
73	Eday Lavas	Shapinsay	197·6	23·2	14°/200	203·5	23·0	7	21·5	13·3°
	Mean, Lavas + contact zone		192·7	44·6	—	—	—	45[2]	7·8	8·2°
			—	—	—	191·3	40·8	45[2]	7·5	8·3°
			194·4	43·4	—	—	—	6[3]	26·9	13·2°
			—	—	—	193·8	40·8	6[3]	20·3	15·2°
22	Eday Marls	S. Ronaldsay	188·4	67·5	10°/240	210·4	74·1	5	30·1	14·1°
			Components with T_B max ≤ 680°C							
17	M. Eday Sst	Deerness	205·3	−4·5	18°/355	205·2	4·6	5	16·0	19·7°
21	U. Eday Sst	Burray	197·9	−31·3	14°/270	196·0	−17·9	4	13·0	26·5°
22	Eday Marls	S. Ronaldsay	177·2	−9·9	10°/240	176·7	−0·9	4	7·9	34·8°
24	M. Eday Sst	S. Ronaldsay	189·1	−25·5	15°/230	184·9	−15·2	13	30·8	16·8°
30	Eday Marls	Kirkwall	187·7	−27·0	13°/170	180·7	−30·2	6	56·0	9·0°
31	Eday Marls	Kirkwall	188·7	−10·4	28°/060	196·0	−31·3	6	24·8	13·7°
	Mean, Eday Group Sediments		190·7	−19·9	—	—	—	38[2]	16·7	5·9°
			—	—	—	189·0	−17·8	38[2]	13·4	6·6°
			190·9	−18·3	—	—	—	6[3]	31·4	12·1°
			—	—	—	190·1	−17·1	6[3]	20·3	15·2°

[1] Dip°/Strike° (dip clockwise from given strike direction)
[2] Unit weight to samples
[3] Unit weight to sites

Downloaded from http://sjg.lyellcollection.org/ at Purdue University Library on July 2, 2015

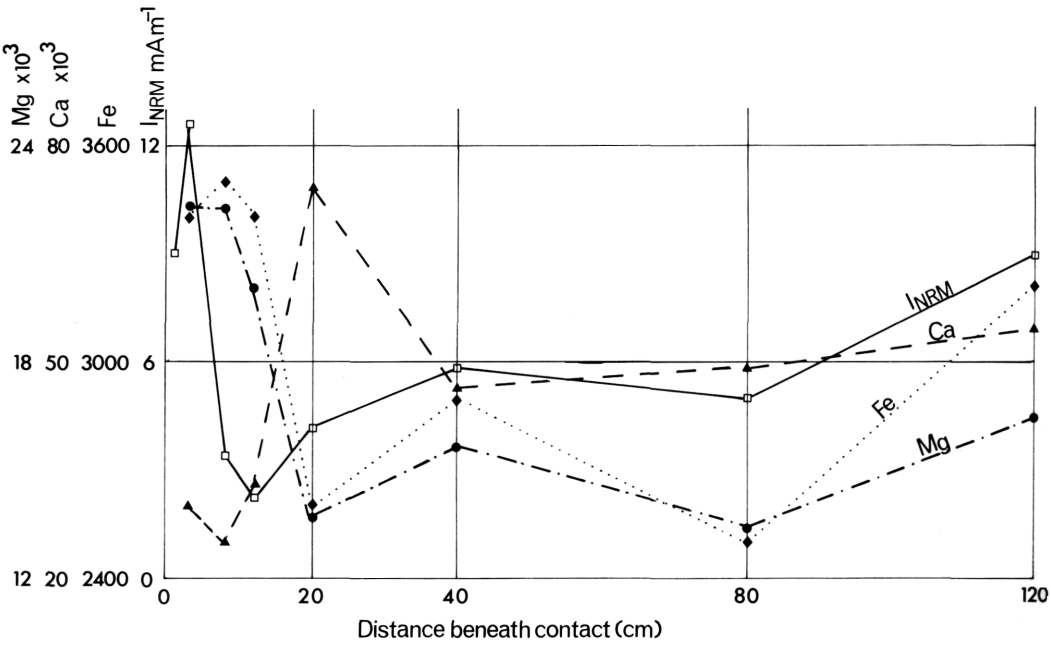


FIG. 4. Major element and NRM intensity variation at site 19 in the Eday Flags beneath the lavas of site 20. Element concentrations (in p.p.m.) measured by I.C.P. spectroscopy.

of the lavas (Table 2). Excluded from the mean are the 5 A.F. results (for the same reasons as for the lavas) and 3 samples showing directions widely separated from the remainder, one of which is aligned with the P.E.F. and two of which are from the same core but show widely disparate directions, suggesting misorientation.

The close agreement between the mean directions in the lava and its contact zone is noteworthy. The extent to which the lava had an observable effect on the sediment is illustrated in Figure 4; this shows that possible iron and magnesium enrichment and calcium depletion have occurred in a zone up to between 12 and 20 cm from the lava; within this zone the flags can be seen to have been baked. There is no increase in NRM intensity beyond 3 cm from the lava, except in the specimens at 1·2 m beneath the contact which also have an anomalously high iron content. Thus, it is not certain whether or not the NRM in the sediment is directly related to the emplacement of the lava. However, given that in places the lava is fairly fresh and free from severe chemical alteration and that the two lithologies are mineralogically very different, the stable NRM directions are unlikely to have a common chemical origin, and there is no evidence in the Orcadian Basin for there ever having been sufficient depth of burial to effect a thermal remagnetisation to the required temperatures. It will be seen from Table 2 that correction of site mean directions for limited local tectonic dip makes no significant difference to the overall mean direction and the variation in the precision parameter k is not

Downloaded from http://sjg.lyellcollection.org/ at Purdue University Library on July 2, 2015

sufficient to give the relative ages of NRM acquisition and deformation (McElhinny 1964).

Taking all the evidence into consideration it seems probable that the mean direction from the lava and its contact zone represents a primary direction, held in magnetite grains, and correlating with the local upper Middle ORS palaeofield direction. It is interesting to note that such a direction may also have been found, but not recognised, by Storetvedt and Petersen (1972, fig. 3); specimens such as OL16-B3 and OL16-C3 show a good grouping around a southerly, moderately steeply downwards direction at temperatures up to the Curie point of magnetite but not above, much in accordance with the results published here, although it is not clear whether such specimens are from Hoy (Upper ORS) or the Eday Lavas.

Eday Group sediments

Lower Eday Sandstone (Sites 11 to 13). Three sites in the Lower Eday Sandstone were collected, two from the Bay of Skaill including one from the coarse-grained yellow sandstones showing well-developed manganese-rich liesegang ring structures, and one site from Taracliff Bay, again with liesegang rings. They all have low NRM intensities (0.1 to 0.8 mA m^{-1}) and are characterised by very low resistance to demagnetisation. There is no consistency between specimens, samples or sites; any semi-stable remanence is considered to reside in iron oxy-hydroxides of indeterminable age. The formation is not considered further.

Middle Eday Sandstone (Sites 16 to 18, 24). Of the four sites in the Middle Eday Sandstone, two (16 and 18) give statistically uninterpretable component distributions although several samples have two peaks in the T_B spectrum, cutting off at approximately 580°C and 680°C. These two remanence components, carried in magnetite and hematite respectively, are in different directions.

The remaining two sites have a stable, high-T_B remanence with a maximum T_B of 675°–685°C, suggesting it to be held in hematite. Blocking temperature ranges for the two sites are somewhat different; site 24 shows little intensity decrease below 600°C (e.g. Specimen 2406B, Fig. 2e) whereas site 17 has a complete T_B range up to 680°C and hence some of the specimens show a distinct low-T_B component, presumably recent in origin (e.g. Specimen 1702, Fig. 2f). Despite this, the two sites give a comparable stable component distribution (Fig. 3b) around a S–SSW shallow mean direction (Table 2).

Eday Marls (Sites 22, 30, 31). Of all the sites analysed, site 22 in the Eday Marls is the only one to show clear, interpretable multicomponent behaviour. This is best illustrated by thermal demagnetisation (e.g. Specimen 2201; Fig. 2g); an inclination change at around 578°C from steeply downwards (at low T_B) to horizontal or slightly upwards is clearly seen. The harder component is not easy to identify in all specimens.

Of the seven specimens analysed, one is rejected because both components are

Downloaded from http://sjg.lyellcollection.org/ at Purdue University Library on July 2, 2015

widely separated from those of the remainder, suggesting misorientation. Five of the remaining six give a low-T_B component (maximum T_B of 570–585°C) directed steeply downwards to the S (Fig. 3c; Table 2). Four give a quantifiable high-T_B component, directed slightly upwards to the south (Fig. 3b, c) as do the other two sites (Fig. 3b, Table 2), neither of which have a comparable low-T_B component.

Upper Eday Sandstone (Site 21). This site, on South Ronaldsay (Fig. 1), is much affected by recent weathering. Several specimens are completely magnetised in a direction approximating to the P.E.F. In the remaining specimens, it is interesting to note that components approximating to the P.E.F. direction are found at both extremes of the blocking temperature spectrum, with both low-T_B and high-T_B components. The former is presumably a VRM, and the latter may be a CRM in a hematitic weathering product. Despite this effect, intermediate T_B-range components are well defined in 5 cores (e.g. specimen 2103, Fig. 2h). These together give a mean directed southerly and upwards (Fig. 3b; Table 2).

Mean directions for the high-T_B components in the Eday Group are given in Table 2. There is no significant difference between mean directions whether taken at site or sample level, although precision is somewhat better at site level. As with the lavas, there is no statistical indication of whether the remanence predates or postdates folding, despite within-site precision being generally greater.

Origin of the remanence

Palaeopole positions corresponding to the mean directions derived in the previous section are compared with an approximate polar wander path (PWP) for NW Europe in Figure 5 (after McElhinny 1973, fig. 112). It can be seen that the pole for the Eday Lavas corresponds most nearly to previously-determined Upper ORS poles, giving strength to the argument that the lavas represent a valid approximation to the upper Middle ORS pole. As the PWP includes no data for the Middle ORS, the result derived in this paper would suggest that post-Silurian westward polar shift was largely completed during the Middle ORS.

The low-T_B component in the Eday Marls, although very poorly-defined as yet, shows a general similarity to comparable results from the lava and its contact, suggesting that the two remanences were acquired at the same time. Examination by electron microscopy of Eday Marls samples from site 22 shows the presence of occasional large (20–100 μ) titanium-bearing iron oxide grains, unequivocally detrital in origin (Fig. 6a). These are unlikely to be titanomaghemite as there is no susceptibility decrease above 350°C, as would be expected with a maghemite-hematite inversion (O'Reilly 1984, p. 16). Hence they are believed to be detrital titanomagnetite grains. Although the observed grains are probably too large to carry a stable ancient remanence (Dunlop 1983), they demonstrate firstly that titanomagnetite grains were deposited at the time of sedimentation and secondly that conditions in the Eday Marls have been suitable for the survival of any smaller

Downloaded from http://sjg.lyellcollection.org/ at Purdue University Library on July 2, 2015

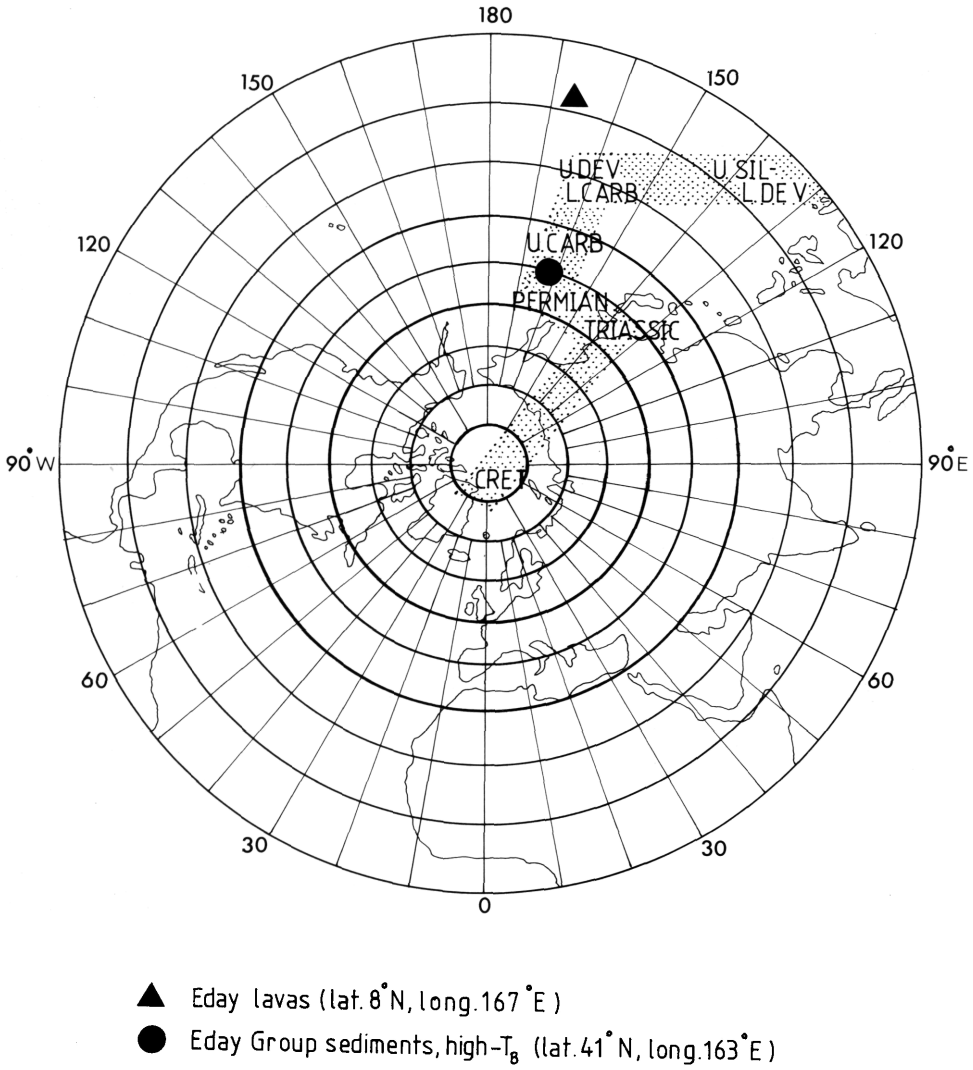


FIG. 5. Generalised polar wander path for NW Europe, Silurian to Cretaceous (after McElhinny 1973). Shows mean poles for the lavas and high-T_B components in Eday Group sediments.

submicroscopic grains which would be able to preserve a primary detrital remanence. It is therefore suggested that the low-T_B component in the Eday Marls represents an upper Middle ORS remanence.

The mean pole for high-T_B components in Eday Group sediments, however, can be seen to lie on the Upper Carboniferous–Lower Permian section of the PWP, suggesting delayed acquisition of a hematitic remanence. As it is everywhere of reversed polarity, it probably corresponds with the Kiaman reversed polarity

Downloaded from http://sjg.lyellcollection.org/ at Purdue University Library on July 2, 2015

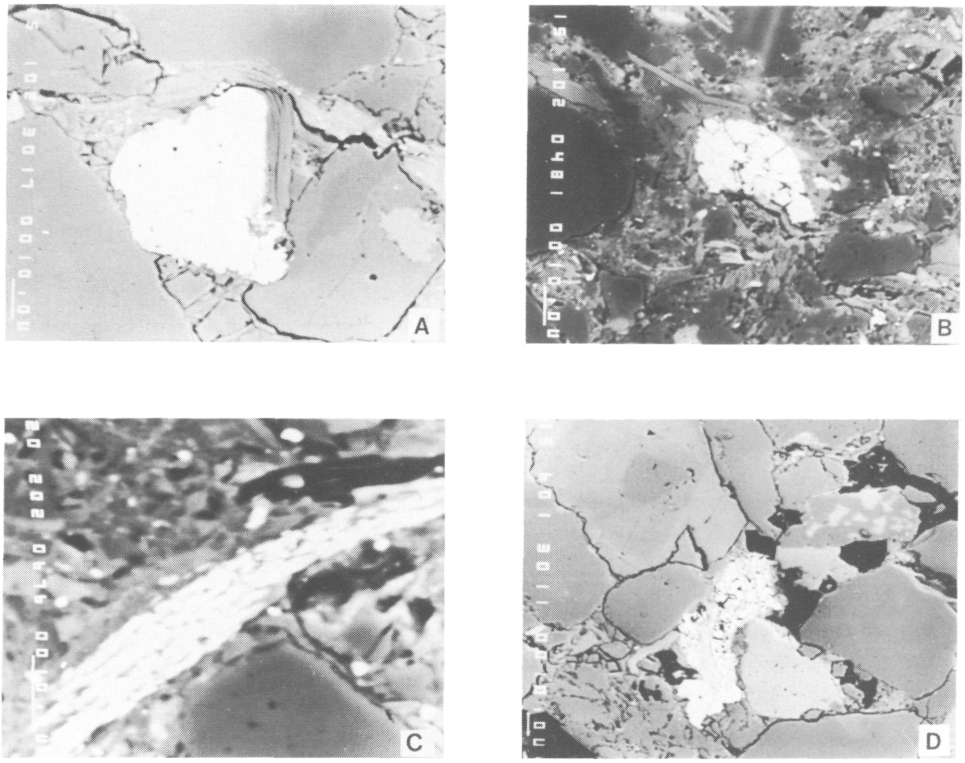


FIG. 6. Backscattered electron images of Eday Group sediments. Scale bar in lower left hand corner is 10 μm. (a) Site 22, Eday Marls; a large (~40 μm diameter) titanium-rich iron oxide grain around which a detrital mica flake has folded. (b) Site 17, Middle Eday Sandstone. Iron oxide grain with a low titanium content (1%), possibly an oxidised magnetite grain. (c) and (d). Site 17. Highly oxidised detrital biotite flakes showing iron oxide exsolution within cleavage planes.

interval. The only evidence seen of late-formed hematite in the Eday Group is the presence of highly martitised detrital iron oxide grains, now fully hematitic (Fig. 6b), and also the widespread occurrence of hematitic exsolution within biotite cleavage planes (Fig. 6c, d). Both of these features have already been observed elsewhere in the Devonian of Northern Scotland (Turner and Archer 1975, 1977). No evidence has been found, to date, of the dedolomitisation of ferroan dolomite as seen elsewhere in the Orcadian Basin (e.g. Sarclet Group, Caithness; Yesnaby Sandstone, Mainland Orkney; John O'Groats Group, Caithness; Robinson, in progress), and which was originally suggested by Tarling (1976) as a possible method of delayed hematite production in carbonate-cemented sandstones in the Orcadian Basin.

All three of these processes (i.e. oxidation of detrital magnetite, oxidation of biotite and dedolomitisation of ferroan dolomite) could have been related to deep

Downloaded from http://sjg.lyellcollection.org/ at Purdue University Library on July 2, 2015

oxidative 'weathering' beneath the Upper Palaeozoic unconformity surface; the pervasiveness and efficiency of such a remagnetisation would be expected to depend greatly on the individual characteristics of a sediment, such as composition, grain size and porosity. This could explain the survival of primary magnetite in fine-grained marls but its complete oxidation in coarser-grained sandstones. The presence or absence of organic matter could also play an important role.

Conclusions

Palaeomagnetic results from a lava and its baked contact have demonstrated the presence of a probable upper Middle Devonian palaeomagnetic direction in these rocks; evidence for this being a primary remanence is based largely on the directional agreement between two very different lithologies. Elsewhere in the Eday Group in the Deerness-South Ronaldsay area, the only formation that appears to have retained a possible syn-depositional or early post-depositional remanence is the Eday Marls, a result so far based on only a small number of specimens. All other sites collected in the Eday Group have given a hematitic remanence, probably of late Palaeozoic to early Mesozoic age, which has replaced or obscured any possible primary remanence in these rocks. The exact cause of this remagnetisation in the Eday Group is not yet proven, but is considered to be one or more of oxidation of primary titanomagnetite grains, oxidation of biotite or dedolomitisation of ferroan dolomite, all of which could be related to deep sub-unconformity weathering in the late Palaeozoic.

Acknowledgements

This paper covers part of the author's PhD research at Leeds University and Imperial College, London, supervised by Professor J. C. Briden and the late Professor J. V. Watson, and aided by a NERC research studentship. Help and assistance from both supervisors at all stages of the work is gratefully acknowledged, as is field assistance by Mr. D. Hatfield. The comments and criticisms of two anonymous referees have been very helpful in revising an earlier version of the paper. Electron microscopy and chemical analyses were performed at Imperial College. The palaeomagnetic studies were made at Leeds University.

References

BRIDEN, J. C., MORRIS, W. A. and PIPER, J. D. A. 1973. Palaeomagnetic studies in the British Caledonides—VI, Regional and global implications. *Geophys. J. R. Astron. Soc.* **34,** 107–34.

——, TURNELL, H. B. and WATTS, D. R. 1984. British palaeomagnetism, Iapetus Ocean and the Great Glen fault. *Geology* **12,** 428–31.

Downloaded from http://sjg.lyellcollection.org/ at Purdue University Library on July 2, 2015

COLLINSON, D. W., CREER, K. M. and RUNCORN, S. K. 1967. *Methods in Palaeomagnetism*. Elsevier, Amsterdam. 609pp.

DUNLOP, D. J. 1983. Determination of domain structure in igneous rocks by alternating field and other methods. *Earth Planet. Sci. Lett.* **63,** 353–67.

KENT, J. T., BRIDEN, J. C. and MARDIA, K. V. 1983. Linear and planar structure in ordered multivariate data, as applied to the progressive demagnetisation of palaeomagnetic remanence. *Geophys. J. R. Astron. Soc.* **75,** 593–622.

McELHINNY, M. W. 1964. Statistical significance of the fold test in palaeomagnetism. *Geophys. J. R. Astron. Soc.* **8,** 338–40.

—— 1973. *Palaeomagnetism and plate tectonics*. Cambridge University Press, 358 pp.

MORRIS, W. A., BRIDEN, J. C., PIPER, J. D. A. and SALLOMY, J. T. 1973. Palaeomagnetic studies of the British Caledonides—IV Miscellaneous new data. *Geophys. J. R. Astron. Soc.* **34,** 69–105.

MYKURA, W. 1976. *British Regional Geology: Orkney and Shetland*. HMSO, Edinburgh.

O'REILLY, W. 1984. *Rock and mineral magnetism*. Blackie, Glasgow. 220pp.

STORETVEDT, K. M. and CARMICHAEL, C. M. 1979. Resolution of superimposed magnetisations in the Devonian John O'Groats Sandstone, North Scotland. *Geophys. J. R. Astron. Soc.* **58,** 769–84.

—— and PETERSEN, N. 1972. Palaeomagnetic properties of the Upper Devonian volcanics of the Orkney Islands. *Earth Planet. Sci. Lett.*, **14,** 269–78.

—— and TORSVIK, T. H. 1983. Palaeomagnetic re-examination of the Basal Caithness Old Red Sandstone; aspects of local and regional tectonics. *Tectonophysics* **98,** 151–64.

TARLING, D. H., DONOVAN, R. N., ABOU-DEEB, J. M. and EL-BATROUK, S. I. 1976. Palaeomagnetic dating of hematite genesis in Orcadian Basin sediments. *Scott. J. Geol.* **12,** 125–34.

TURNER, P. and ARCHER, R. 1975. Magnetisation history of Lower Old Red Sandstones from the Gamrie Outlier, Scotland. *Earth Planet. Sci. Lett.*, **27,** 240–250.

—— and —— 1977. The role of biotite in the diagenesis of red beds from the Devonian of Northern Scotland. *Sed. Geol.* **19,** 241–51.

WILSON, G. V., EDWARDS, W., KNOX, J. JONES, R. C. B. and STEVENS, J. V. 1935. The Geology of the Orkneys. *Mem. Geol. Surv.*

ZIJDERFELD, J. D. A. 1967. A.C. demagnetisation of rocks: analysis of results. *In* Collinson, D. W., Creer, K. M. and Runcorn, S. K. (eds). *Methods in Palaeomagnetism*. Elsevier, Amsterdam: 609pp.

MS. accepted for publication 31*st May* 1985